

9.8.4 RCC PLL configuration register (RCC_PLLCFGR)

Address offset: 0x00C

Reset value: 0x0000 1000

Access: no wait state, word, half-word and byte access

This register is used to configure the PLL clock outputs according to the formulas:

- $f(\text{VCO clock}) = f(\text{PLL clock input}) \times (\text{PLLN} / \text{PLLM})$
- $f(\text{PLL_P}) = f(\text{VCO clock}) / \text{PLL P}$
- $f(\text{PLL_Q}) = f(\text{VCO clock}) / \text{PLL Q}$
- $f(\text{PLL_R}) = f(\text{VCO clock}) / \text{PLL R}$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PLLPDIV[4:0]					PLLR[1:0]		PLLREN	Res.	PLLQ[1:0]		PLLQEN	Res.	Res.	PLL P	PLLPEN
rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw			rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	PLLN[6:0]							PLLM[3:0]				Res.	Res.	PLLSRC[1:0]	
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw			rw	rw

Bits 31:27 **PLLPDIV[4:0]**: Main PLL division factor for PLLSAI3CLK

Set and cleared by software to control the SAI1 or SAI2 clock frequency.

PLLSAI3CLK output clock frequency = VCO frequency / PLLPDIV.

00000: PLLSAI3CLK is controlled by the bit PLLP

00001: Reserved

00010: PLLSAI3CLK = VCO / 2

....

11111: PLLSAI3CLK = VCO / 31

Bits 26:25 **PLLR[1:0]**: Main PLL division factor for PLLCLK (system clock)

Set and cleared by software to control the frequency of the main PLL output clock PLLCLK. This output can be selected as system clock. These bits can be written only if PLL is disabled.

PLLCLK output clock frequency = VCO frequency / PLLR with PLLR = 2, 4, 6, or 8

00: PLLR = 2

01: PLLR = 4

10: PLLR = 6

11: PLLR = 8

Caution: The software must set these bits correctly not to exceed 110 MHz on this domain.

Bit 24 **PLLREN**: Main PLL PLLCLK output enable

Set and reset by software to enable the PLLCLK output of the main PLL (used as system clock).

This bit cannot be written when PLLCLK output of the PLL is used as system clock.

In order to save power, when the PLLCLK output of the PLL is not used, the value of PLLREN must be 0.

0: PLLCLK output disabled

1: PLLCLK output enabled